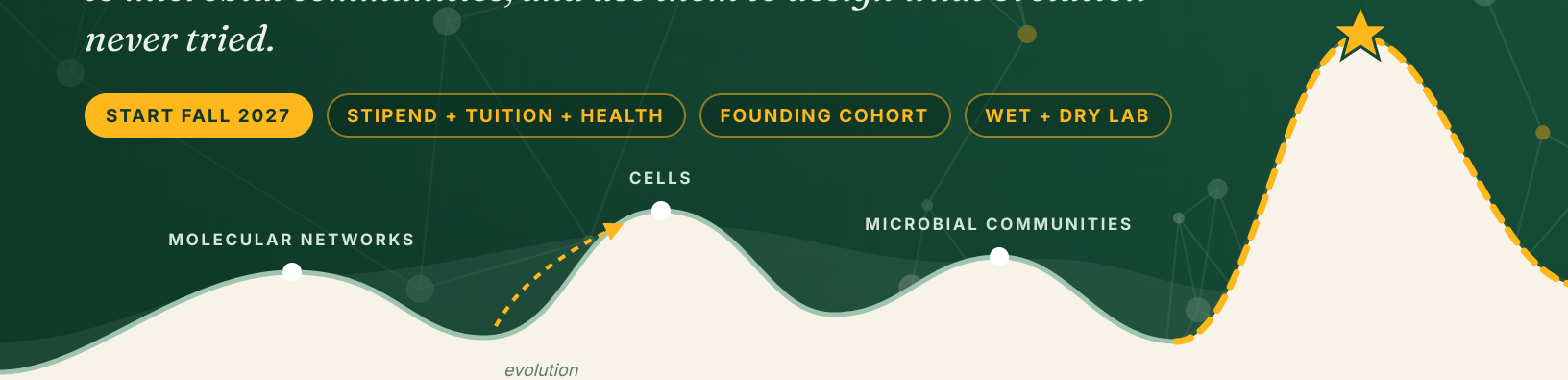
BIOJAZZ LAB · DEPARTMENT OF BIOLOGY · BAYLOR UNIVERSITY · WACO, TX

1–3 Fully Funded *PhD Positions*

Uncover the **design principles of life**, *from molecular networks to microbial communities, and use them to design what evolution never tried.*

START FALL 2027 | STIPEND + TUITION + HEALTH | FOUNDING COHORT | WET + DRY LAB



Why are living systems built the way they are, and *how can we design new ones?*

We use microbes as tractable, fast-evolving model systems. We pair **multi-omics** (transcriptomics, proteomics, metabolomics, and other phenomics) with **mechanistic yet data-constrained virtual cell models**, and use AI and automation to speed up the whole cycle from model to experiment.

01 Cell Physiology & Communities

How cells allocate limited resources, and how those choices shape microbial communities.

02 Predictive Phenomics

Predict how cells respond to genetic, environmental, and drug perturbations.

03 Evolutionary Systems & Synthetic Biology

Evolve cellular networks *in silico* with BioJazz, then design beyond evolution.

04 AI for Biology

AI tools linking proteomics to protein structure and function, at scale.

IMPACT Aging & Disease | Infection & Immunity | Bioeconomy | One Health

SKILLS YOU'LL BUILD

- **Multi-omics** experiments & data analysis
- **Virtual cell models**: mechanistic, data-constrained
- ***In silico* evolution** & rule-based modeling
- **AI, automation & scientific software**

Curiosity matters more than background.

Biology | Physics | Chemistry | Math | CS | Engineering

WHAT WE OFFER

Full funding
Stipend, tuition, and health insurance throughout the PhD.

Founding member
Shape the lab's projects, tools, and culture from day one.

Close mentorship
One-on-one training with the PI in a small, supportive team.

Wide network
Clinical & national-lab partners; PI formerly at PNNL & LANL.

APPLY BY DECEMBER 1, 2026 · START FALL 2027



Email Dr. Song Feng before you apply

Send a short note on your interests and your CV, then apply to the **Baylor Biology PhD program** and name the BioJazz Lab in your application.

Song_Feng@baylor.edu biojazz-lab.github.io/join

SCAN FOR DETAILS

RECENT WORK IN *Science Advances* • *PRX Life* • *Protein Science* • *Cell Reports Methods* • *Nucleic Acids Research*